CYBV312
Python Scripting in Cybersecurity

Week 7

Extracting Meta Data from JPEG



Week Seven Agenda

- Review the Week 6 Solution
- The Python Image Library (3rd Party)

Week Seven Assignments

Your To Do List

- Attend: Week Seven Live Lecture
- Quiz 7: Using PIL
- Assignment 6: Creating a Photo Search Script

Key Ideas

Topics

- What information can we glean from JPEG Meta Data using the Python Image library (PIL)

Thinking more deeply about classes/ objects and library creation: Targeting cybersecurity

- Key questions:
  - What cybersecurity challenges lend themselves to library creation?
  - What factors should we consider?
  - How can we share/create our own Python Libraries

Key Cybersecurity Challenges

- A few possibilities
  - Image and Multimedia Analysis
  - Digital Forensic Incident Response (DFIR)
  - Tripwire
  - Deception Methods
  - NSRL Processing
  - Targeted Machine Learning
    - Malware Analysis
    - Indications of Compromise (IOC) Search
    - Natural Language Interpretation



Considerations

- Key Factors
  - What is the key challenge we would like to focus on and how is it solved today?
  - Does a library already exist? If yes, can we extend or leverage the existing library(s)?
  - Can we integrate our solution into an existing cybersecurity technology or tool?
  - How might this benefit the community?



Sharing

Sharing Options

- Python Package Index: https://pypi.org/
- Github: https://github.com/
- Conference Presentation
- Short Book
- Blog, Youtube



Processing images with Python Image Library

Installation

NOTE:From the Windows Command Line or MAC/Linux Terminal Window

- Make sure to first ensure that pip is the latest version
  - **python -m pip install --upgrade pip**
- Then install the library
  - **python -m pip install --upgrade Pillow**
- Special Note:
  - If you have multiple versions of python installed (i.e. 2.x and 3.x) then make sure to specify the appropriate Python interpreter. For example:
    - **python3 –m install –upgrade Pillow**
- **Library Handbook**
  - https://pillow.readthedocs.io/en/latest/handbook/index.html



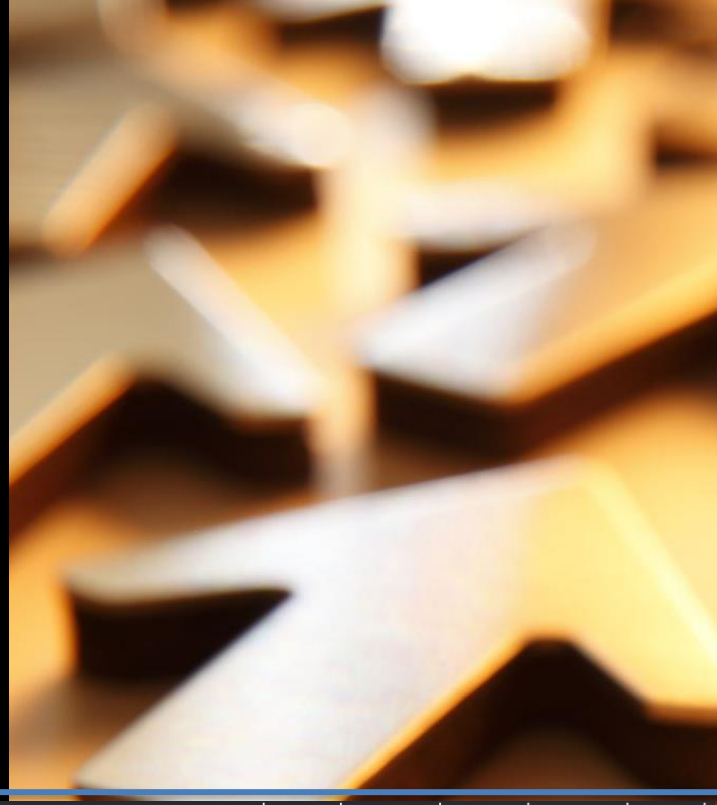
Searching for Photos by content

```
'''
Searching for Images with PIL
Professor Hosmer
August 2020
'''
import sys
import os

from PIL import Image
from prettytable import PrettyTable

tbl = PrettyTable(['File', 'Ext', 'Format', 'Width', 'Height', 'Mode'])
DIR = ".\\photos"
fileList = os.listdir(DIR)
for eachFile in fileList:
    path = os.path.join(DIR, eachFile)
    if os.path.isfile(path):
        ext = os.path.splitext(path)[1]
        try:
            with Image.open(path) as im:
                tbl.add_row([path, ext, im.format, im.size[0], im.size[1], im.mode])
        except Exception as err:
            tbl.add_row([path, ext, "[NA]", "[NA]", "[NA]", "[NA]"])
            pass
    else:
        continue

tbl.align = 'l'
print(tbl.get_string(sortby="Format"))
```



File	Ext	Format	Width	Height	Mode
.\photos\PH01236U.BMP	.BMP	BMP	216	143	P
.\photos\PH02039U.BMP	.BMP	BMP	216	143	P
.\photos\PH02752U.BMP	.BMP	BMP	216	142	P
.\photos\38467giu.gif	.gif	GIF	300	212	P
.\photos\AG00004_.GIF	.GIF	GIF	140	135	P
.\photos\AG00011_.GIF	.GIF	GIF	131	148	P
.\photos\AG00021_.GIF	.GIF	GIF	109	150	P
.\photos\AG00037_.GIF	.GIF	GIF	114	150	P
.\photos\Biking.jpg	.jpg	JPEG	1136	852	RGB
.\photos\Cat.jpg	.jpg	JPEG	1280	853	RGB
.\photos\CoastLine.JPG	.JPG	JPEG	4032	3024	RGB
.\photos\Deutchland.JPG	.JPG	JPEG	1600	1200	RGB
.\photos\Disney.jpg	.jpg	JPEG	1408	940	RGB
.\photos\Farm.jpg	.jpg	JPEG	625	469	RGB
.\photos\Munich.JPG	.JPG	JPEG	1200	1600	RGB
.\photos\Turtle.jpg	.jpg	JPEG	800	533	RGB
.\photos\zzz.jpg	.jpg	JPEG	225	225	RGB
.\photos\Avatar.png	.png	PNG	222	378	RGBA
.\photos\PacketHackingVillage.png	.png	PNG	300	301	P
.\photos\Photo-2.png	.png	PNG	428	300	RGBA
.\photos\monalisa.png	.png	PNG	310	432	RGBA
.\photos\Digital Signatures Paper.doc	.doc	[NA]	[NA]	[NA]	[NA]
.\photos\First Four Papers.doc	.doc	[NA]	[NA]	[NA]	[NA]

Extracting JPEG EXIF Data with PIL

```
'''
Python Image Library Basics
Extracting EXIF Data from a JPEG
Professor Hosmer
August 2020
'''
import sys

# pip install Pillow
from PIL import Image
from PIL.ExifTags import TAGS, GPSTAGS

# Extract Basic EXIF Data
print("\nExtract JPEG EXIF Data \n")
imageFileName = input("Enter JPEG: ")

try:
    pilImage = Image.open(imageFileName)
    exifData = pilImage._getexif()

except Exception as err:
    sys.exit("\nPIL Exception: "+str(err))

if not exifData:
    sys.exit("\nNo EXIF data\n")

# Process eachTag
for tag, theValue in exifData.items():

    tagValue = TAGS.get(tag, tag)
    tagData  = exifData.get(tag)

    if tagValue != "GPSInfo":
        # Display the Basic Tag Data
        print("="*40)
        print("TAG:  ", tagValue)
        print("Data: ", tagData)
    else:
        print("GPS Data Found")
        # Process GPS data
        for tag in theValue:
            gpsTag = GPSTAGS.get(tag, tag)
            print("\t", gpsTag, theValue[tag])
```

```
Extract JPEG EXIF Data

Enter JPEG: castle.jpg
========================================
TAG:   ExifVersion
Data:  b'0320'
========================================
TAG:   ComponentsConfiguration
Data:  b'\x00\x03\x02\x01'
========================================
TAG:   FlashPixVersion
Data:  b'0010'
========================================
TAG:   DateTimeOriginal
Data:  2012:06:09 12:42:24
========================================
TAG:   DateTimeDigitized
Data:  2012:06:09 12:42:24
========================================
TAG:   ExposureBiasValue
Data:  0.0
========================================
TAG:   ColorSpace
Data:  1
========================================
TAG:   MeteringMode
Data:  2
========================================
TAG:   ExifImageWidth
Data:  1280
========================================
TAG:   Flash
Data:  16
----------------------------------------
```

```
========================================
TAG:   FocalLength
Data:  18.0
========================================
TAG:   ExifImageHeight
Data:  848
========================================
TAG:   ExifInteroperabilityOffset
Data:  54723
========================================
TAG:   FocalLengthIn35mmFilm
Data:  27
========================================
TAG:   SceneCaptureType
Data:  0
========================================
TAG:   Contrast
Data:  0
========================================
TAG:   Saturation
Data:  0
========================================
TAG:   Sharpness
Data:  2
========================================
TAG:   SubjectDistanceRange
Data:  3
========================================
TAG:   Make
Data:  PENTAX
========================================
TAG:   Model
Data:  PENTAX K-5
========================================
TAG:   Orientation
Data:  1
```

```
========================================
TAG:   YCbCrPositioning
Data:  2
========================================
TAG:   SensingMethod
Data:  2
========================================
TAG:   Copyright
Data:  Steen Gyes Bruun
========================================
TAG:   FileSource
Data:  b'x'
========================================
TAG:   ExposureTime
Data:  0.002
========================================
TAG:   XResolution
Data:  72.0
========================================
TAG:   YResolution
Data:  72.0
========================================
TAG:   FNumber
Data:  8.0
========================================
TAG:   SceneType
Data:  b'i'
========================================
TAG:   ExposureProgram
Data:  2
```

```
GPS Data Found
        GPSVersionID b'\x00\x00\x03\x02'
        GPSLatitudeRef \000
        GPSLatitude (55.0, 0.4403, 0.0)
        GPSLongitudeRef \000
        GPSLongitude (11.0, 54.656, 0.0)
        GPSAltitudeRef b'\x00'
        GPSAltitude 15.9
        GPSTimeStamp (11.0, 42.0, 25.0)
        GPSSatellites PAD\0009
        GPSStatus \000
        GPSMeasureMode \000
        GPSSpeedRef \000
        GPSSpeed 1.01
        GPSTrackRef \000
        GPSTrack 122.03
        GPSImgDirectionRef \000
        GPSImgDirection 112.03
        GPSMapDatum WGS-84
        GPSProcessingMethod b'ASCII\x00\x00\x00GPS'
        GPSDateStamp 2012:06:09
```

Assignment

Photo Search

This week you will be generating a photo search script leveraging this weeks briefing, demonstration and sample scripts.

1) You will prompt the user for a starting folder (use the zip folder provided to create your test directory)

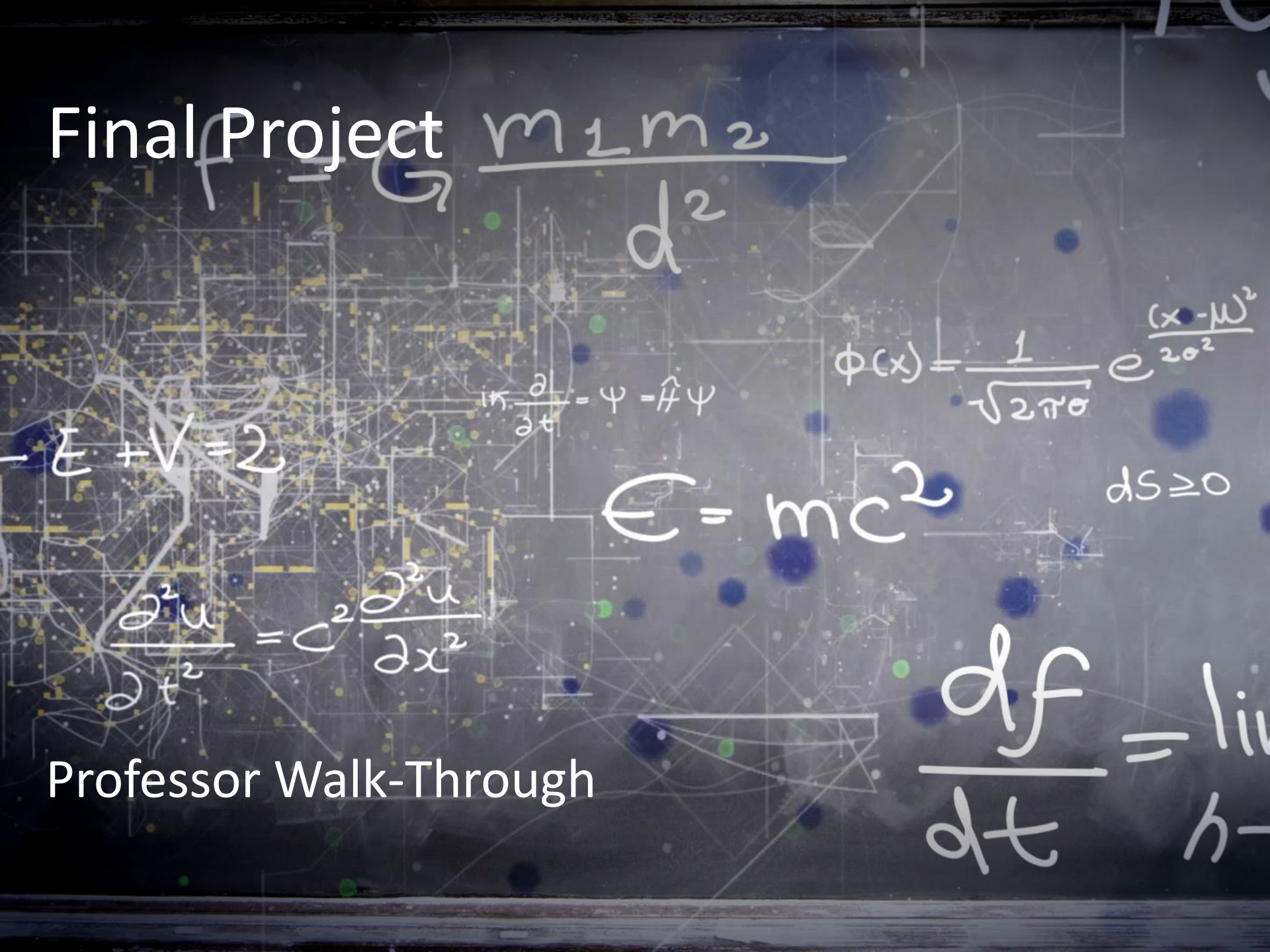
2) You will examine each file and determine if it is an image (using the PIL library method), and then extract and report the path, extension, format, width, height and mode for each.

3) Your script will produce a Prettytable the same as the SearchPhotos.py script provided in D2L.

4) You will submit:

- Your python script
- A screenshot of the resulting prettytable

Final Project



Professor Walk-Through